PERFECT COMPLEXES WITH SUPPORT LIFT WITH A SHIFT: F PLUS F[1] EXTENDS OVER THE AMBIENT AFFINE SCHEME

THE STACKS PROJECT AUTHORS

Editorial note. Input lemma-lift-perfect-complex-plus-shift is same-chapter NOT carded (cross-line-claimed label) and is declared external.

Allowed standard prerequisites (editorial).

- Killing maps by Koszul powers. lemma-perfect-into-support-on-T (same chapter, carded at reading position, see specs). After replacing g_j by powers, multiplication by g_j is zero on F.
- Lifting perfect complexes over affines. lemma-lift-perfect-complex-plus-shift (same chapter, NOT carded, declared external). Provides E on X with E|_U = F and the g_j acting nilpotently.
- Cone of multiplication by a Koszul power. Derived Categories, lemmas split and third-map-square-zero (cross-chapter, named). C_1|_U = F + F[1] + F[1] + F[2] and g_1 squares to zero on C_1.
- Lifting maps by tensoring with perfect complexes. lemma-lift-map (same chapter, carded). Provides beta, gamma: C' -> C_s with C' = C_s tensor^L I and prescribed restrictions.
- Pushforward of supported perfect complexes. Cohomology, lemma pushforward-perfect (cross-chapter, named). Keeps the intermediate objects perfect and supported on T.

Source reference. Original reference 05QT: focused-seeds/geometry/S9-perf2-lift-perfect-complex-plus-shift-support/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 05QP: focused-seeds/geometry/S9-perf2-lift-perfect-complex-plus-shift-support/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 08EH: sources/stacks-perfect/pinned-original-source.tex, source lines 3571–3571. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 08EI: sources/stacks-perfect/pinned-original-source.tex, source lines 3615–3615. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 08EJ: sources/stacks-perfect/pinned-original-source.tex, source lines 3644–3644. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let X be an affine scheme. Let $T \subset X$ be a closed subset such that $X \setminus T$ is quasi-compact. Let $U \subset X$ be a quasi-compact open. For every perfect object F of $D(\mathcal{O}_U)$ supported on $T \cap U$ the object $F \oplus F[1]$ is the restriction of a perfect object E of $D(\mathcal{O}_X)$ supported in T.*

Proof. Say $T = V(g_1, \ldots, g_s)$. After replacing g_j by a power we may assume multiplication by g_j is zero on F, see Lemma 08EJ. Choose E as in Lemma 08EI. Note that $g_j : E \to E$ restricts to zero on U. Choose a distinguished triangle

$$E \xrightarrow{g_1} E \to C_1 \to E[1]$$

By Derived Categories, Lemma 05QT the object C_1 restricts to $F \oplus F[1] \oplus F[1] \oplus F[2]$ on U. Moreover, $g_1 : C_1 \to C_1$ has square zero by Derived Categories, Lemma 05QP. Namely, the diagram

$$\begin{array}{ccccc} E & \longrightarrow & C_1 & \longrightarrow & E[1] \\ {\scriptstyle 0}\downarrow & & {\scriptstyle g_1}\downarrow & & {\scriptstyle 0}\downarrow \\ E & \longrightarrow & C_1 & \longrightarrow & E[1] \end{array}$$

is commutative since the compositions $E \xrightarrow{g_1} E \to C_1$ and $C_1 \to E[1] \xrightarrow{g_1} E[1]$ are zero. Continuing, setting C_{i+1} equal to the cone of the map $g_i : C_i \to C_i$ we obtain a perfect complex C_s on X supported on T whose restriction to U gives

$$F \oplus F[1]^{\oplus s} \oplus F[2]^{\oplus \binom{s}{2}} \oplus \ldots \oplus F[s]$$

Choose morphisms of perfect complexes $\beta : C' \to C_s$ and $\gamma : C' \to C_s$ as in Lemma 08EH such that $\beta|_U$ is an isomorphism and such that $\gamma|_U \circ \beta|_U^{-1}$ is the morphism

$$F \oplus F[1]^{\oplus s} \oplus F[2]^{\oplus \binom{s}{2}} \oplus \ldots \oplus F[s] \to F \oplus F[1]^{\oplus s} \oplus F[2]^{\oplus \binom{s}{2}} \oplus \ldots \oplus F[s]$$

which is the identity on all summands except for F where it is zero. By Lemma 08EH we also have $C' = C_s \otimes^{\mathbf{L}} I$ for some perfect complex I on X. Hence the nullity of $g_j^2 \mathrm{id}_{C_s}$ implies the same thing for C'. Thus C' is supported on T as well. Then $\mathrm{Cone}(\gamma)$ is a solution. $\square$